

TOPIC OUTLINE

1. Steps in Diagnosis of Neurologic Disease 1.1 Series of Steps that Consist of Clinical Method 1.2 Major Categories of Neurologic Disease 1.3 Case 1	3.3 The Comatose Patient 3.4 Psychiatric Patient 3.5 Infants of Small Children 3.6 General Medical Examination 3.7 Others
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3. Neurologic Examination 3.1 Patients Represents with Symptoms of Nervous System Disease 3.2 The Medical or Surgical Patient without Neurologic Symptoms	5. Case Assignment 5.1 Case 2
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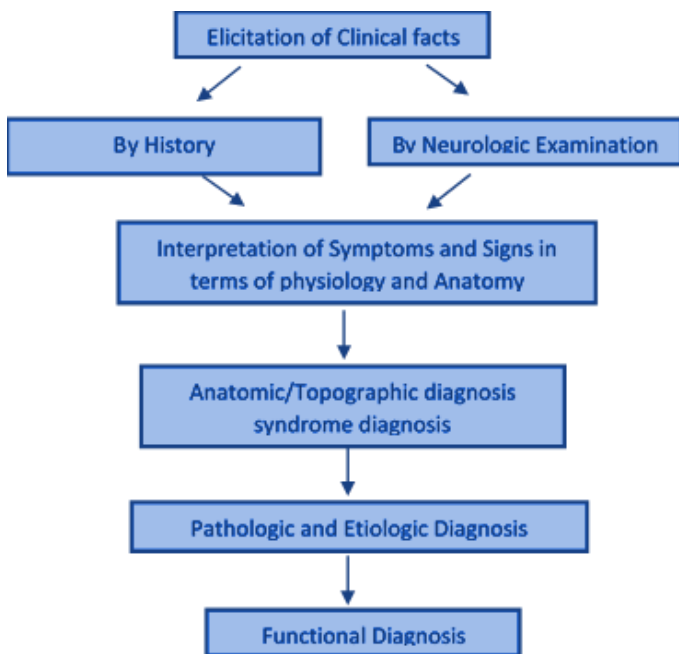
1 STEPS IN DIAGNOSIS OF NEUROLOGIC DISEASE

Figure 1. Steps in Diagnosis of Neurologic Disease

1.1 SERIES OF STEPS THAT CONSISTS OF CLINICAL METHOD

- **Elicitation of Clinical Facts**
 - By taking complete and thorough history, physical and complete neurologic examination
- **Interpretation of Symptoms and Signs in Terms of Physiology and Anatomy**
 - Example: Sudden onset of hemiplegia/hemiparesis → left-sided (**Physiologic Diagnosis**)
 - Interpret based on the history
 - Determine whether the cause is acute, subacute or chronic
 - **Acute:** Vascular diseases (stroke) – most common, Trauma (sudden onset of focal lesions)
 - **Chronic:** Infection, Space-occupying lesions, Autoimmune
 - Elicit sign and symptoms to rule in and rule out autoimmune disease
 - Example: Left-sided weakness including the face, arms and legs (**Anatomic Diagnosis**)
 - **Lateralization:** Right side of the NS
 - **Level:** Cerebrum

- **Localization:** Medulla
 - Tract involved: Corticospinal tract
 - **Medulla** contains the ARAS (responsible for wakefulness) and CN IX, X, XI, XII
- Example: Patient is awake, can answer questions but has slurred speech, with left central facial palsy, pupils are equal and briskly reactive to light, there is weakness on both upper and lower extremities (1/5), (+) Babinski reflex (L), can move eyeballs in all directions (**Anatomic Diagnosis**)
 - **Localization:** Basal Ganglia
 - Dense hemiplegia with intact sensorium
 - Contains the internal capsule

Nice to Know!

- **Know the level of the lesion**
 - Is it in the cerebral cortex, brainstem, cerebellum, spinal cord, peripheral nerves, and neuromuscular junction or in the muscles?
- **Know the lateralization of the lesion**
 - Is it on the right, left or midline?
- **Localize the lesion - specific area**
 - Importance of knowing the localization of the lesion: to know what tests to request (because you should not request cranial CT scan if lesion is in the muscle)
 - Example: If lesion is in the brainstem or spinal cord, instead request a cranial MRI because it is surrounded by bones. If in the peripheral nerves, request EMG NCV.
 - TB meningitis is usually found in the brainstem or base of the brain. Usually bilateral rectal palsy, on-and-off fever for several weeks, nuchal rigidity, our initial diagnosis is TB meningitis especially if the patient is Filipino.
 - **History + Neurologic Examination** makes 99% of the diagnosis
 - **Gerstmann Syndrome:** right-left confusion and inability to write, calculate and identify individual fingers, lesion located in the region of the left angular gyrus

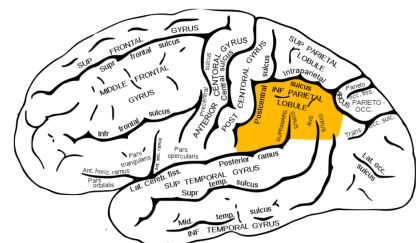


Figure 2. Location of the angular gyrus

Anatomic/ Topographic Diagnosis, Syndromic Diagnosis

- Gives syndromic diagnosis
 - Example: Weber's syndrome
- **Pathologic and Etiologic Diagnosis**
 - By having the anatomic/topographic / syndromic diagnosis, pathologic and etiologic diagnosis can be obtained
- **Functional Diagnosis**
 - How long the patient should stay in the hospital depends on the cause of the disease

1.2 MAJOR CATEGORIES OF NEUROLOGIC DISEASE

- Infectious
- Genetic – congenital
- Traumatic
- Degenerative
- Vascular
- Toxic
- Metabolic (acquired or Inherited)
- Neoplastic
- Inflammatory – Immune

- Psychogenic
- Iatrogenic

must be obtained from a spouse, relative, friend, or employer

1.3 CASE 1

- **Patient is awake; Has weakness of the left face, left arm, and left leg; Decreased sensation of the left side of the body; (+) Babinski on the left side.**
 - **Level:** Cerebral hemisphere
 - Could it be possible that the lesion is in the spinal cord?
 - Patients with spinal cord lesion can have both motor and sensory deficits and can present with (+) Babinski sign (especially if the lesion is in the cervical area which involves the tract before the anterior horn cells exit from the spinal cord)
 - But most often in the spinal cord lesion you can seldom have hemiparesis because spinal cord has a very small tendency to have a motor neuron lesion -> anterior horn cell -> lower motor neuron (-) Babinski; flaccidity instead of spasticity
 - Why not in the peripheral nerve?
 - Because you cannot find hemi-sensory manifestation in peripheral nerves -> individual nerves (Specific location of innervation)
 - Multiple nerves can be involved (polyneuropathies) but most often it is bilateral not hemi. The same is true with your muscles.
 - Lateralization: Right side
 - Localization: Medulla
 - Medulla contains the ARAS (responsible for wakefulness) and CN IX, X, XI, XII
 - Midbrain: CN III and IV
- **Pathologic Diagnosis**
 - **Case:** Patient is playing basketball and suddenly complains of headache, dizziness and left sided weakness. What could be the pathologic diagnosis of the patient?
 - **Aneurysm/Stroke** (vascular in etiology)
 - If **chronic/progressive**, it can be a tumor/ space occupying lesion
- **Functional Diagnosis**
 - How long the patient should stay in the hospital depends on the cause of the disease
- **Prevalence and Incidence of Neurologic Diseases**
 - ★ **Stroke: Most commonly encountered condition**

2 HISTORY TAKING

- Needs patient's trust and cooperation
- Avoid suggesting to the patient
 - If you're thinking of a peripheral nerve disease, do not ask the patient "May nabatagan ikaw nga daw gatuslok-tuslok?" Instead, ask what the patient feels
 - Ask open-ended questions
 - For example, the patient complains of headache. Let the patient describe the type of headache he/she experiences. But if a patient can't really describe, that's the time you can suggest.
 - Errors and inconsistencies in the recorded history are as often the fault of the physician as of the patient. The patient should be discouraged from framing his symptom(s) in terms of a diagnosis that he may have heard; rather, he should be urged to give as **accurate a description of the symptom as possible**.
 - Elicit the setting in which illness occurred, mode of onset and evolution
 - When asking questions, you can backtrack because sometimes they only note when the symptom is already severe
 - As neurologic diseases often impair mental function, it is necessary in every patient who might have cerebral disease for the physician to decide whether the patient is competent to give a history of the illness
 - If the patient's powers of attention, memory, and coherence of thinking are inadequate, the history

3 NEUROLOGIC EXAMINATION

3.1 PATIENTS WHO PRESENTS WITH SYMPTOMS OF NERVOUS SYSTEM DISEASE

- Testing of **Higher Cortical Functions**
- Testing for **Cranial Nerves**
- Test of **Motor Function**
- Test of **Reflex Function**
- Test of **Sensory Function**
- Testing of **Gait and Stance**

TESTING OF HIGHER CORTICAL FUNCTIONS

- ★ **Psychiatric Aspects** which incorporate:
 - Affect
 - Mood
 - and Normality of thought processes and content
- ★ **Neurologic Aspects** which include:
 - Level of consciousness
 - Awareness (attention)
 - Language
 - Memory
 - and visuospatial abilities
- Questions are first **directed toward** determining the patient's **orientation in time and place** and **insight** into his current medical problem
- **Attention, Speed of response, ability to give relevant answers to simple questions and capacity for sustained and coherent mental effort** all lend themselves to straightforward observation
- **Bedside tests**
 - For attention, concentration, memory and clarity of thinking
 - Includes:
 - Repetition of a series of digits in forward and reverse order
 - Serial subtraction of 3s, or 7s from 100
 - and recall of three items of information
 - Short story after an interval of 3 minutes
- **Excellent Tests for Memory** includes:
 - Patient's account of his recent illness
 - Dates of hospitalization
 - Day-to-day recollection of recent incidents
- **Test for language ability and Coherence of Thinking**
 - Narration of the illness
 - Patient's choice of words (vocabulary)
- If there is any suggestion of a **speech or language disorder**:
 - The nature of the patient's **spontaneous speech** should be **noted**
- The accuracy of reading, writing, and spelling, executing spoken commands, repeating words and phrases spoken by the examiner, naming objects and parts of objects and solving simple arithmetical problems should be assessed.
- **Test for visuospatial perception**
 - Indicated in cases of *suspected cerebral disease*
 - Includes:
 - Bisecting a line
 - Drawing a clock
 - Floor plan of one's home or map of one's country
 - Copying figures

3.2 THE MEDICAL OR SURGICAL PATIENT WITHOUT NEUROLOGIC SYMPTOMS

- Orientation, insight, language

- ★ **Insight** - very important in differentiating a neurologic patient from a psychiatric patient
- Size and reaction of light of pupils, visual and auditory acuity
 - If the patient's pupil is dilated and sluggishly reactive
 - You cannot say immediately that the patient has neurologic problem
 - It can be caused by intoxication to medication (Diazepam)
- Movement of eyes, face, tongue
- Examination of the outstretched hands for atrophy, pronating or downward drift, tremor, power of grip and wrist dorsiflexion
 - Early atrophy** of the hands occur at the **lower motor neuron** diseases
- Biceps, supinator, and triceps tendon reflexes
- Inspection of the legs during active flexion and extension of hips, knees and feet
- Patellar, Achilles and Plantar reflexes
- Vibration sensibility in the fingers and toes
- Finger-to-nose and Heel-to-shin testing
 - For coordination
 - Test for cerebellar function
- Gait

3.3 THE COMATOSE PATIENT

- ★ **Glasgow Coma Scale**
 - To localize lesion
 - Includes:
 - Eye Response
 - Verbal Response
 - Motor Response
 - Signs of Focal Irritation
 - Meningeal irritation
 - Increased ICP
 - (+) Papilledema
 - Metabolic Diseases

Glasgow Coma Scale					
EYE OPENING		VERBAL RESPONSE		MOTOR RESPONSE	
					
Spontaneous	4	Oriented	5	Obey commands	6
To sound	3	Confused	4	Localizing	5
To pressure	2	Words	3	Normal flexion	4
None	1	Sounds	2	Abnormal flexion	3
		None	1	Extension	2
GLASGOW COMA SCALE SCORE					
Mild 13-15		Moderate 9-12		Severe 3-8	

Figure 3. Glasgow Coma Scale and Scoring

Nice to Know!

Signs of Meningeal Irritation

- Presence of **Brudzinski's** and **Kernig Sign**
- Brudzinski's Sign**
 - Severe neck stiffness causes a patient's hips and knees to flex when the neck is flexed
- Kernig Sign**
 - (+) test if the leg extension causes pain

3.4 PSYCHIATRIC PATIENT

- Important to differentiate neurologic patient from psychiatric patient

- ★ **In psychiatric patient:**
 - Auditory hallucination** - most common symptom
 - Hallucination is vivid
- ★ **In Neurologic patient**
 - Hallucination is not vivid or rather it is blurry hallucination**
- Depends on cooperation of the patient
 - Example: Anxious and malingering patients
 - You cannot really examine the patient properly
 - You have to gain the trust and confidence of the patient
 - Psych patients sometimes are very manipulative so be careful

3.5 INFANTS OR SMALL CHILDREN

- Just observe the patient because they cannot still speak

3.6 GENERAL MEDICAL EXAMINATION

- Underlying systemic diseases that has secondarily affected the nervous system
- The neurological manifestations of thyrotoxicosis include tremors and, in severe cases, coma
- If with meningeal signs, do lumbar puncture

3.6 OTHERS

- IMPORTANCE OF WORKING KNOWLEDGE OF NEUROANATOMY, NEUROPHYSIOLOGY AND NEUROPATHOLOGY**
- LABORATORY DIAGNOSIS**
- THERAPEUTICS IN NEUROLOGY**

Nice to Know!

- The neurologic examination begins with observation of the patient while the history is being obtained
- The manner in which the patient tells the story of his illness may betray confusion or incoherence in thinking, impairment of memory or judgement, or difficulty in comprehending or expressing ideas
- Asking the patient to give his own interpretation of the possible meaning of symptoms may sometimes expose unnatural concern, anxiety, suspiciousness, or even delusional thinking
- Then proceeds to an examination of the cranial nerves, neck, and trunk to the testing of motor, reflex, and sensory functions of the upper and lower limbs
- This is followed by an assessment of the function of sphincters and the autonomic nervous system and testing for meningeal irritation by examining the suppleness of the neck and spine
- Gait and station (standing position) should be observed before or after the rest of the examination
- If there is decreased sensorium, check for blood sugar
- The examination must also be modified according to the condition of the patient

4 CLINICAL METHODS

4.1 SHORTCOMINGS OF THE CLINICAL METHOD OF NEUROLOGY

- Focus on the clinical analysis of the principal signs and symptoms and should avoid being distracted by minor signs and uncertain clinical data
- Avoid Early Closure of Diagnosis
- When several of the main features of the disease are lacking, alternative diagnosis should always be entertained

4. It is preferable to base one's experience with the dominant symptoms and signs and not on statistical analyses of clinical phenomena
5. Obtain tissue for examination if possible

FOCUS ON THE CLINICAL ANALYSIS OF THE PRINCIPAL SIGNS AND SYMPTOMS AND SHOULD AVOID BEING DISTRACTED BY MINOR SIGNS AND UNCERTAIN CLINICAL DATA

- Some minor signs may be a major contributing factor to the case of the patient
- Example:
 - If the patient complains of weakness of both lower extremities and ascending paralysis and says that it's occasional and mild "nga daw ga tuslok-tuslok sang nag ligad pa nga anom ka bulan tapos na dula naman"
 - DO NOT DISREGARD OR TAKE FOR GRANTED
 - Consider GBS or multiple sclerosis in which tingling sensation can be one manifestation and other peripheral nerve diseases
 - If a patient has a headache, focus on the headache. If patient has allergies, focus on allergies

AVOID EARLY CLOSURE OF DIAGNOSIS

- Do not automatically assume a final diagnosis without considering other possible diagnoses
- There may other symptoms in the past other than the sudden cause of the patient's disease
 - Example:
 - The patient was out and about day prior, then sudden onset of left sided weakness, do not assume stroke right away.
 - From the patient's history, he had fever 3 weeks ago, so it could be viral encephalitis with fever and behavioral changes which resolved then the patient had a stroke; it is not solely vascular in origin, viral encephalitis must be identified that occurred few weeks ago

THE PATIENT WAS OUT AND ABOUT DAY PRIOR, THEN SUDDEN ONSET OF LEFT SIDED WEAKNESS, DO NOT ASSUME STROKE RIGHT AWAY

- For example, in GBS, patient has:
 - Ascending paralysis
 - Flaccidity
 - (-) Babinski
 - (+) Fibrillations
 - Fasciculation
 - **Generalized weakness** is also characterized by ascending paralysis but (-) fasciculation, (-) hypotonia
 - We should also consider other disease such as:
 - Multiple sclerosis
 - Muscle diseases (myopathies)

IT IS PREFERABLE TO BASE ONE'S EXPERIENCE WITH THE DOMINANT SYMPTOMS AND SIGNS AND NOT ON STATISTICAL ANALYSES OF CLINICAL PHENOMENA

- If in doubt give more importance to the patient's signs and symptoms and not the clinical data
- Example: Multiple sclerosis is more common in females.
 - If a male patient comes in with signs and symptoms of multiple sclerosis, do not rule out MS just because of his gender

OBTAIN TISSUE FOR EXAMINATION IF POSSIBLE

- If the patient has intracranial lesions, the patient must undergo a pathologic examination of the mass (Craniotomy or removal of the lesion) or if you are considering that the patient has Hansen's disease, take a sample of the nerve to examine whether the patient has the disease or not

4.2 PURPOSE OF CLINICAL METHOD OF NEUROLOGY

- Enables the physician to determine proper treatment of the patient
- Helpful in determining the prognosis
- Allows of genetic Counseling for hereditary diseases
- It is the essential initial step in the scientific study of clinical phenomena and disease

5 CASE ASSIGNMENT

5.1 F.P., a 23 year old, right handed female, Roman Catholic, from Jaro, Iloilo City, admitted for the first time.

- **Chief Complaint:** difficulty in swallowing
- **History of Present Illness**
 - 2 weeks PTC (2 weeks to 6 mos subacute disease)
 - (+) Fever
 - Relieved by *paracetamol*
 - 1 week PTC
 - (+) **numbness of both lower** extremities
 - No weakness noted
 - No consult
 - 2 days PTC
 - (+) **heaviness of both lower** extremities
 - 1 day PTC
 - (+) difficulty in ambulating
 - (+) difficulty in opening mouth and raising eyebrows
 - (+) difficulty in swallowing
 - (+) weakness of both legs and arms
- **Personal and Social History:**
 - Student
 - Non-smoker, non-alcoholic beverage drinker
- **Family History**
 - (+) HTN - both parents
- **Physical Examination:** No significant findings
- **Neuro Examination:**
 - **MSE**
 - Conscious, coherent, oriented, intact memory, cooperative, wheelchair borne, in mild respiratory distress
 - **Cranial Nerves**
 - I - able to smell coffee
 - II - distinct disc margins, no hemorrhages, no visual field out, VA: 20/20
 - II, III - intact pupillary light reflex
 - III, IV, VI - intact EOM movements
 - VII - cannot wrinkle forehead, unable to puff nor purse lips
 - VIII - normal Rinne's and Weber's test
 - IX, X - weak gag reflex
 - XI - can shrug shoulder
 - XII - tongue midline on protrusion, no atrophy, no fasciculations
 - **Motors:**
 - (+) pronator drift between upper extremities
 - % both LE
 - **Sensory:** Intact
 - **Cerebellar functions**
 - Can perform finger-to-nose test
 - Heel-to-knee-to-shin test with slight difficulty

- **Deep Tendon Reflexes:** (+)
- No Babinski

Interpretation:

- **★ Lesion is in the brainstem**
 - As evidenced by CN VII, IX, and X assessment findings which are found in the brainstem
 - **Corticospinal tract is affected**
 - more severe weakness is observed in the lower extremities
 - Which also travels through the brainstem
- Classified as **★ Lower motor neuron problem**
 - As evidenced by **hyporeflexia** and **absence of Babinski reflex**
- **★ Weakness + sensation loss** = lesion is in the peripheral nerve
- **★ Weakness only** - lesion might be in peripheral nerve, muscle or neuromuscular junction
- **Neuromuscular junction lesion**
 - Weakness is absent in the morning and after resting periods because the neurotransmitter acetylcholine is released from the neuromuscular junction
- **EMG NCV**
 - To check whether the problem is in the muscles or in the nerves

Supplemental Information:

Source: O'Sullivan & Schmitz's Physical Rehabilitation, 8th Edition

- This is a case of GBS which is a lower motor neuron lesion. Note that the initial event which is a fever followed by sensory complaints starting from the lower extremity. This was further solidified by the gradual loss of motor function.
- While GBS is a self-limiting condition, the symptoms can sometimes be severe enough to require a patient mechanical ventilator due to respiratory distress.

Nice to Know!

Acute Disease - 1 day to 2 weeks

Subacute Disease - 2 weeks to 6 months

Chronic Disease - 6 months or longer

6 REFERENCES

- Adams and Victor's Principles of Neurology 11th Edition
- Dr Tamos' Powerpoint presentation and discussion
- OmVi Trans

7 PRACTICE QUESTIONS

- Which of the following can contribute to 99% of the diagnosis in neurology?**
 - CT scan and MRI
 - Lumbar puncture and biopsy
 - History and neurologic examination
 - Genetic testing and toxicology screen
- A 60-year-old patient presents with sudden-onset right hemiplegia. What is the most likely etiology if the onset is acute?**
 - Tumor
 - Stroke
 - Multiple sclerosis
 - Guillain-Barré syndrome
- Dense left hemiplegia with preserved sensorium localizes the lesion most likely to which of the following structures?**
 - Basal ganglia

- Spinal cord
- Brainstem
- Internal capsule

- Which of the following findings is most consistent with a lower motor neuron lesion?**
 - Hyperreflexia and Babinski sign
 - Hyporeflexia and absent Babinski sign
 - Clonus and spasticity
 - Extensor plantar reflex and increased tone
- What clinical sign is present when flexing the neck causes involuntary flexion of the hips and knees?**
 - Kernig's sign
 - Hoffmann's sign
 - Brudzinski's sign
 - Babinski sign
- A patient presents with slurred speech, facial asymmetry, and absent gag reflex. Which cranial nerves are likely involved?**
 - V, VI, VII
 - VII, IX, X
 - IX, XI, XII
 - III, IV, VI
- A patient presents with right-sided hemiparesis and left-sided cranial nerve palsy. The lesion most likely localizes to the:**
 - Cerebral cortex
 - Basal ganglia
 - Brainstem
 - Spinal cord
- Which of the following best distinguishes a psychiatric patient from a neurologic patient?**
 - Positive Babinski reflex
 - Impaired visuospatial skills
 - Vivid auditory hallucinations
 - Flaccid paralysis
- A 25-year-old male with bilateral muscle weakness has normal brain imaging. What is the next best diagnostic test if a muscle disorder is suspected?**
 - Cranial CT scan
 - MRI of the spinal cord
 - EMG and nerve conduction studies
 - CSF analysis
- A patient has right-left confusion, finger agnosia, agraphia, and acalculia. What syndrome does this suggest?**
 - Wallenberg syndrome
 - Gerstmann syndrome
 - Weber syndrome
 - Brown-Sequard syndrome

Answers: CBDBC BCCCB

